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RESEARCH

Dasatinib boosts γδ T cell expansion and memory phenotypes with enhanced antitumor immunity

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Abstract

Γδ T cells offer unique advantages in cancer immunotherapy because of their MHC-independent recognition of tumor antigens and innate cytotoxic potential. However, conventional ex vivo expansion protocols using zoledronic acid (Zol) and IL-2 often lead to terminal differentiation and diminished effector function. In this study, we performed a flow cytometry-based screen of an FDA-approved compound library to identify agents that enhance γδ T cell expansion while preserving their functional phenotypes. Dasatinib, a clinically used tyrosine kinase inhibitor, has emerged as a promising therapeutic candidate. Dasatinib-treated Vδ2 T cells (γδ2 T-Da) exhibited increased expansion, elevated expression of memory-associated markers (CD62L and CD127), reduced apoptosis, and higher production of TNF-α and IFN-γ. Transcriptomic analysis revealed the upregulation of genes related to T cell survival and self-renewal, including MYC, TCF7, and MIR155HG. Functionally, γδ2 T-Da cells demonstrated superior cytotoxicity against glioblastoma (GBM) and triple-negative breast cancer (TNBC) cells in vitro with sustained activity after prolonged culture. In orthotopic tumor models, γδ2 T-Da cells enhanced tumor control, reduced TNBC metastasis, and prolonged survival of GBM-bearing mice. These results suggest that dasatinib improves γδ T cell yield and function, providing a practical and translatable strategy for optimizing γδ T cell-based adoptive therapy, particularly for solid tumors.

Trial registration: Trial number: CMUH111-REC3-185

Keywords γδ T cells · Dasatinib · Drug screening · Ex vivo expansion · Glioblastoma · Triple-negative breast cancer

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Introduction

Γδ T cells have emerged as promising candidates for adoptive cell therapy in cancer treatment, offering several advantages over conventional αβ T cell-based immunotherapies. Unlike αβ T cells, γδ T cells recognize target cells independently of major histocompatibility complex (MHC) molecules, rendering them resistant to common tumor immune evasion strategies, such as β2-microglobulin loss and neoantigen depletion [1–3]. Their T cell receptors (TCRs) recognize a wide range of monomorphic stress-induced antigens that are frequently overexpressed across various tumor types, enabling broad MHC-independent tumor targeting [4, 5].

In addition to TCR-mediated recognition, γδ T cells express a range of natural killer (NK) cell receptors, endowing them with NK-like "normality sensing" abilities. This allows them to discriminate malignant cells from healthy tissue, contributing to their antitumor activity through direct cytotoxicity, support of CD8 + T cell memory, and tissue repair [6–8]. Furthermore, γδ T cells, particularly human Vγ9Vδ2 subset, can function as antigen-presenting cells (APCs). They can express MHC class II and costimulatory molecules, allowing them to present antigens and activate other T cells to induce immune responses [9, 10]. These unique characteristics position γδ T cells as a versatile and potent platform for adoptive cell therapy, whether using unmodified or genetically engineered approaches.

Ex vivo expansion of γδ T cells from peripheral blood mononuclear cells (PBMCs) has been widely studied. In humans, the predominant circulating γδ T cell subsets are Vγ9Vδ2 and Vγ(x)Vδ1. A major breakthrough occurred in 1999 when Vγ9Vδ2 T cells (also termed γδ2 T cells) robustly proliferated in response to aminobisphosphonates, such as pamidronate [11]. This expansion is now known to involve Vδ2 TCR recognition of mevalonate pathway-derived phosphoantigens (pAgs) presented by BTN2A1/3A1 dimers, an evolutionarily conserved immune-sensing system present in primate [12–14]. Based on this mechanism, zoledronic acid (Zol), combined with cytokines such as IL-2 or IL-15, is now commonly used in clinical protocols for γδ T cell expansion [15, 16].

Extended ex vivo culture is often necessary to achieve sufficient cell numbers for clinical application. However, prolonged culture may lead to cellular exhaustion, impaired memory phenotypes, and reduced antitumor functions [17–19]. Therefore, preserving γδ T cell functionality during expansion is critical for therapeutic success. In this study, we first sought to characterize the phenotypic changes that occur during γδ T cell expansion using accessible flow cytometry and transcriptomic methods. Given the complexity of γδ T cell differentiation and activation networks, we employed an unbiased high-throughput screen of FDA-approved compounds to identify agents that could enhance γδ T cell expansion and functionality. Drug repurposing provides the advantage of leveraging clinically validated compounds with established safety profiles, thereby accelerating their clinical translation and reducing development costs [20]. Using this approach, along with GMP-grade reagents, we identified unexpected compounds and molecular pathways that significantly improved γδ T cell proliferation, memory phenotypes, and antitumor activity. These findings open new avenues for directly translating γδ T cell-based immunotherapy strategies from bench to clinical applications.

Materials and methods

Reagents and antibodies

All the antibodies used in this study are listed in supplementary Table S1, including their species, clones, catalog numbers, and applications. The oligonucleotide primers used for quantitative real-time PCR (qPCR) are listed in supplementary Table S2. Primers were designed based on published sequences or generated using Primer-BLAST and synthesized by Genomics BioSci & Tech Co., Ltd. (New Taipei City, Taiwan). A full list of reagents and chemical compounds is summarized in supplementary Table S3, including key experimental materials such as compounds, recombinant cytokines, apoptosis detection kits, and staining reagents. All reagents were obtained from commercial sources and used according to the manufacturer's instructions unless otherwise specified.

Detailed protocols for RNA sequencing and pathway analysis, qPCR, caspase-3/7 activity assays, statistical analysis, and co-culture cytotoxicity assays are described in the Supplementary Materials and Methods.

Ethics declaration

The human samples used in this study adhered to the principles outlined in the Declaration of Helsinki and conformed to all relevant ethical guidelines for biomedical research. All data analyses involved patient samples in accordance with institutional regulations and relevant ethical norms, as approved by the Ethical Committees of China Medical University Hospital, Taichung, Taiwan (CMUH111-REC3-185), and all donors provided written informed consent prior to tissue collection.

All animal studies followed the National Institutes of Health Guidelines for Animal Research and were approved by the Animal Research Center of the National Chung Hsing

University, Taichung, Taiwan (IACUC No. 111–128). This study was designed, conducted, and reported in accordance with the ARRIVE 2.0 guidelines for animal research.

Study period

Experiments and data collection for human PBMC samples were performed from September 2022 to November 2024, and for animal models from March 2023 to April 2025.

Expand γδ2 cells from PBMC

Zol-based cell expansion was performed according to previously established protocols [15, 16]. Briefly, PBMC (1×10^6/ml) from healthy donors were seeded in U-shaped 96-well plates. The culture medium contained optimizer CTS T cell expansion medium (A1048501, Thermo Fisher) support with 5% human platelet lysate (UltraGRO Advanced), 5 µM zoledronic acid (MedChemExpress), and 200 IU/ml recombinant human IL-2 (PeproTech, 200–02) was added into PBMCs. During the cell expansion, half of the medium was refreshed every 2 days. Information about the healthy blood donors is listed in supplementary Table S4. Tyrosine kinase inhibitors (TKIs) were purchased from Cayman Chemical, including imatinib (no. 13139), dasatinib (no. 11498), bosutinib (no. 12030), ponatinib (no. 11494), and nilotinib (no. 10010422). Cryopreserved PBMCs were used only in animal experiments when required by experimental scheduling constraints. PBMCs were cryopreserved in CELLBANKER® 1 (ZENOGEN PHARMA) according to the manufacturer's instructions. Upon thawing, PBMCs were rested for 48 h in IL-2-supplemented medium (100 IU/mL), centrifuged to remove cellular debris, and then used to initiate γδ T cell expansion.

FDA-approved drug library

During Zol-based cell expansion, each drug from the FDA-approved drug library (Cayman, Item No. 23538) was added to culture medium at day 3. Half of the medium containing the drugs was refreshed every 2 days, the cell density was maintained between 1×10^6/ml to 5×10^6/ml. The expansion efficiency and the expression of surface markers were analyzed by flow cytometer.

Flow cytometry data analysis

Γδ2 T cells were harvested from human peripheral blood mononuclear cells (PBMC) and gated on CD3+CD56+single cells. Prior to intracellular cytokine staining (ICS), cells were restimulated for 5 h in the presence of PMA (50 ng/ml), ionomycin (500 ng/ml), and Brefeldin A (10ug/ml). The cell surface markers were labeled first, and intracellular proteins were stained with fluorochrome-conjugated antibodies after fixation and permeabilization. Flow cytometry data were analyzed using FlowJo, SPICE v6.1 [21], and Pestle v2.0.

Clustering analysis and advanced data visualization on days 0, 3, 5, 7, and 10 γδ2 T cells were performed using FlowJo plugins (FlowSOM, t-SNE, and ClusterExplorer). FlowSOM was employed to generate nine distinct subclusters, which were then dimensionally reduced using t-SNE and visualized using ClusterExplorer. For SPICE analysis, Boolean gating analysis was performed on γδ2 T cells to examine the combinatorial expression patterns of five cytokines (TNF, IFN, IL2, IL17A, and IL4). From the 32 subsets, pie chart representations revealed the ability of dasatinib to augment the proportion of cytokine secretion by γδ2 cells in normal and cancer cell co-culture environments. Individual and overlapping arcs surrounding the pie chart represent marker-positive cells for each subset.

Cell culture

Human glioblastoma cell lines Ln18 were purchased from the American Type Culture Collection (Manassas, VA); human glioblastoma cell lines GBM8901 and human breast adenocarcinoma cell line MDA-MB231 were purchased from Bioresources Collection and Research Center (Food Industry Research and Development Institute, Hsinchu, Taiwan). Cells were cultured in DMEM (Life Technologies, Grand Island, NY, USA) with 10% fetal bovine serum (FBS), 2 mM glutamine, 100 U/ml penicillin, and 100 µg/ml streptomycin. All cell cultures were maintained at 37 °C in 5% CO2 in a humidified incubator.

In vivo experiments

All tumor-bearing mouse models were established using 6-week-old female NOD-SCID mice (National Laboratory Animal Center, Tainan, Taiwan). For the orthotopic breast cancer model, 10^6 MDA-MB-231 cells were injected into the third mammary fat pad. For the glioblastoma model, 10^5 GBM8901 cells were injected intracranially into the right cerebral hemisphere using stereotactic apparatus (right, 2 mm; backward, 1 mm; depth, 3 mm). During surgery, gas anesthesia with isoflurane was administered, and pain medication was administered postoperatively. The animal experimental design adhered to the 3R principles (Replacement, Reduction, and Refinement). If the mouse's weight fell below 20% of its original weight, it was humanely euthanized using carbon dioxide. Human γδ T cells were isolated and cultured for ex vivo expansion on day 7 for adoptive cell therapy. In the breast cancer model, tumors were allowed to grow for 5 weeks, and γδ T cells (2×10^6) were administered via intratumoral injection twice a week for 3 weeks. In the glioblastoma model, γδ T cells (2×10^6) were injected into

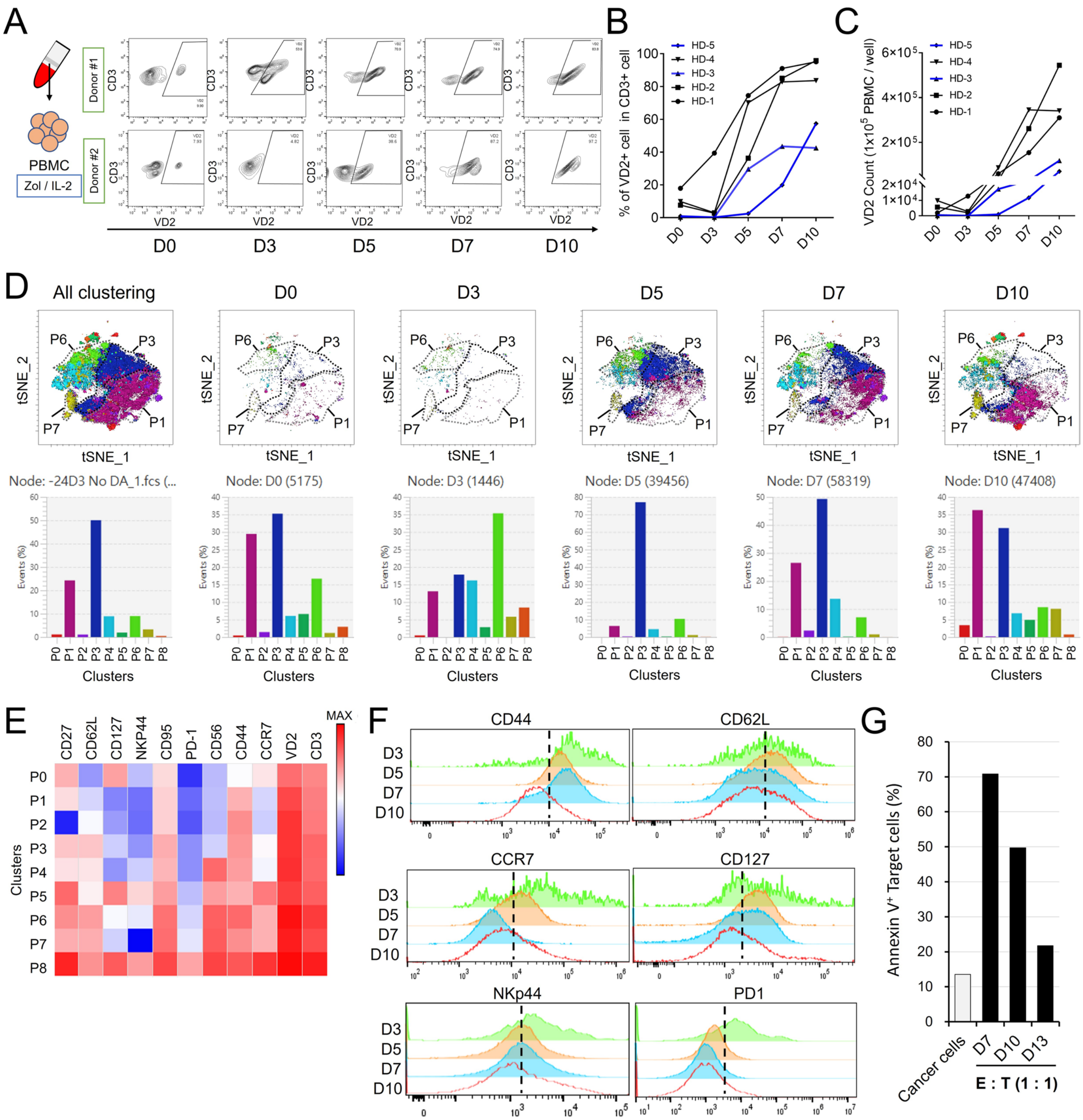


Fig. 1 Phenotypic characterization of ex vivo expanded γδ2 T cells. **A** Overview of Zol-based γδ2 T cell expansion from human PBMCs. **B**–**C** Time-course analysis of γδ2 cell population dynamics from five healthy donors (HD-1 to HD-5), showing initial decline followed by robust expansion. The blue line indicates the poor expander (HD-3 and HD-5), whose γδ2 T cells constitute less than 60% of all CD3^{+} cells. **D** t-SNE and FlowSOM clustering identified nine phenotypically distinct CD3^{+}VD2^{+}-γδ2 T cell subsets (P0–P8) during cell expansion based on the expression of surface markers from the pool sample of five healthy donors. **E** Marker expression heatmap defines γδ2 T cell subsets (P0–P8). **F** Histogram analysis of key memory and cytotoxic markers over culture duration, data from the pool sample of five healthy donors. **G** Cytotoxicity of γδ2 T cells from HD-1 against Ln18 cells decreases with extended culture, as measured by co-culture apoptosis assays

the retro-orbital sinus. In both models, zoledronic acid (Zol, 200 μg/kg) was administered intraperitoneally 1 day prior to γδ T cell injection.

For the cell tracking experiments, control cells were labeled with CFSE (BioLegend, Cat. No. 423801), whereas dasatinib-expanded cells were labeled with BioTracker™

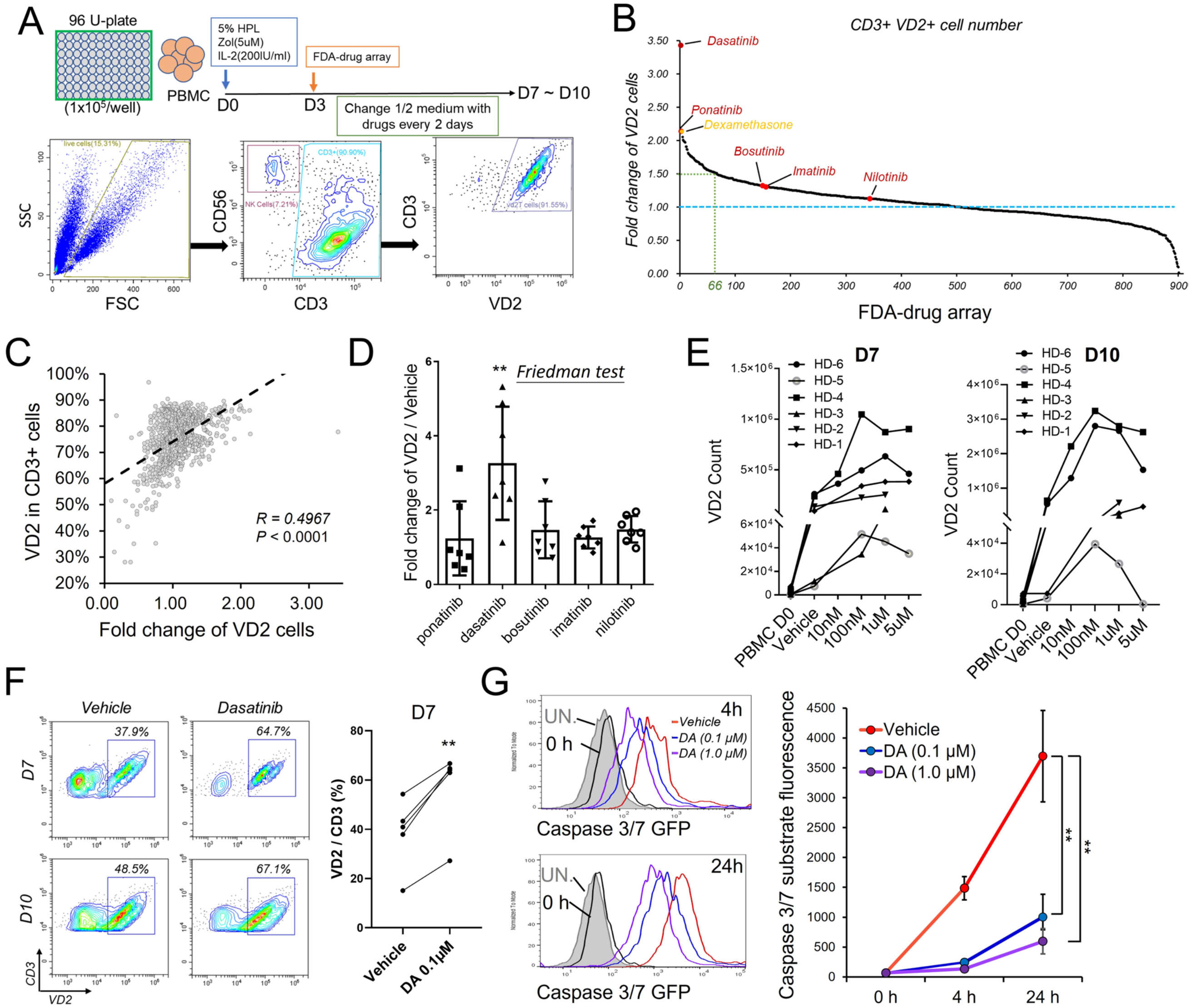


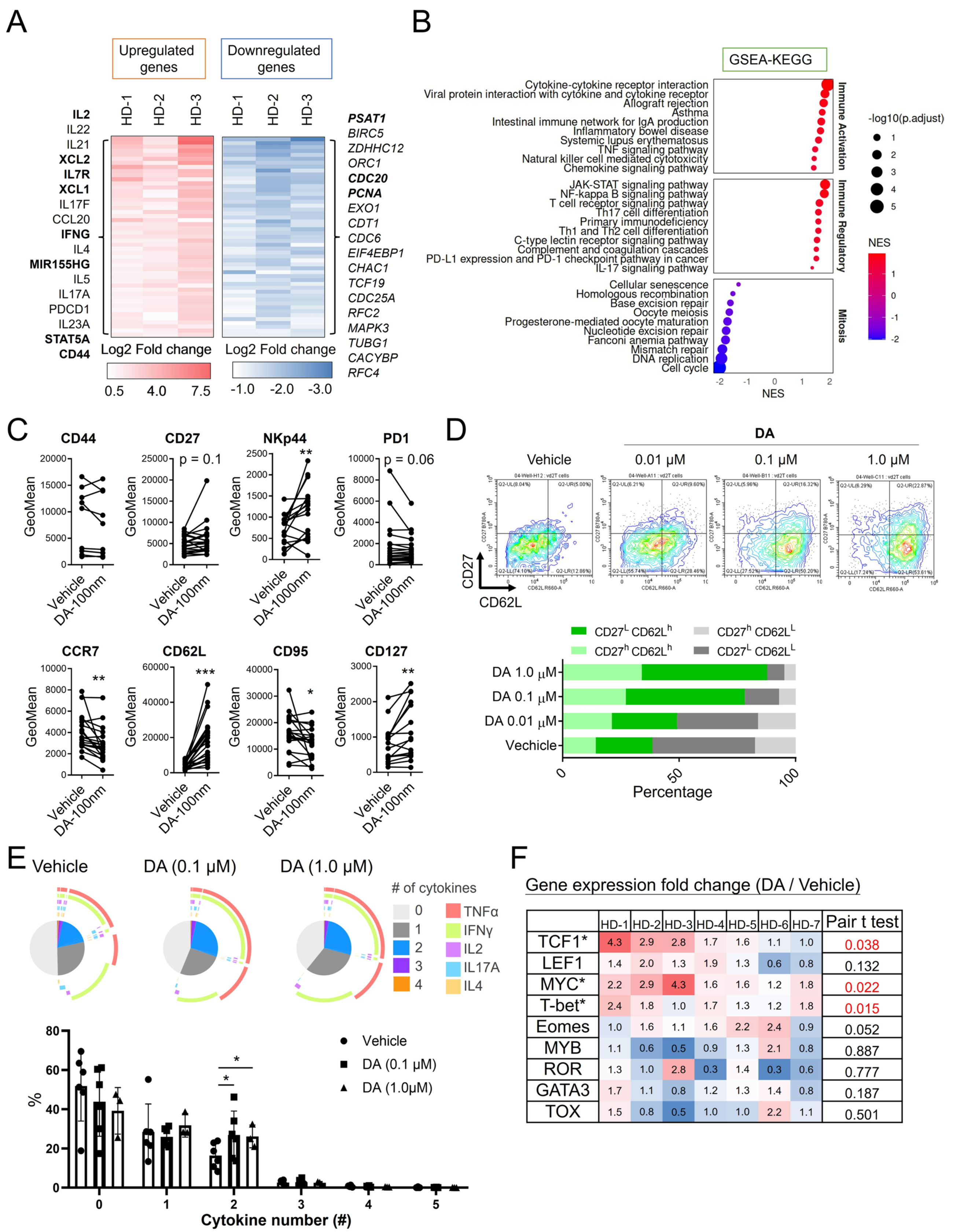
Fig. 2 Drug screening identifies dasatinib as a top enhancer of γδ2 T cell expansion. **A** Screening and cell gating strategy using an FDA-approved drug library during Zol-IL2 culture. **B** Fold-change comparison of γδ2 cell expansion across two donors, highlighting dasatinib and ponatinib as top candidates. Average fold change of γδ2 cell number refers to vehicle control (blue dash line). Sixty-six compounds revealed that 50% increases of γδ2 cell number were indicated. **C** Association of γδ2 cell expansion and percentage of γδ2 cell in total T cell population (CD3$^+$). **D** Validation of five TKIs; dasatinib shows strongest expansion in multiple donors. Data are presented as the mean ± SD from seven healthy donors. Friedman test is used, $^{**}p<0.01$. **E** Dose–response curve of dasatinib in γδ2 expansion. (F) Comparison of γδ2 T cell expansion in poor expanders with or without dasatinib. Paired *t*-test is used, $^{**}p<0.01$. (G) Caspase-3/7 activity analysis indicating dasatinib suppresses apoptosis. Results are presented as means ± SD. Student *t*-test is used, $^{**}p<0.01$

655 Red Cytoplasmic Membrane Dye (Millipore, Cat. No. SCT087). 2×10^6 of labeled cells were mixed and injected into glioblastoma-bearing NOD-scid mice via the retro-orbital sinus. Mice were sacrificed 24, 48, and 96-h post-injection, and the following tissues were harvested: bone marrow, lymph nodes, spleen, peripheral blood, tumor-injected brain regions, and adjacent normal brain tissues. The distribution of CFSE$^+$ and 655$^+$ γδ T cells in each tissue was analyzed by flow cytometry.

Results

Phenotypic characterization of ex vivo expanded γδ2 T cells from human PBMCs

Circulating γδ T cells comprise a mixed population. Although an individual's γδ T cell status is relatively stable over time after birth [22], its changes during ex vivo expansion remain poorly understood. The Zol-based expansion method is widely used to effectively promote γδ2 T


A
Upregulated genes
Downregulated genes
HD-1 HD-2 HD-3
HD-1 HD-2 HD-3
IL2
IL22
IL21
XCL2
IL7R
XCL1
IL17F
CCL20
IFNG
IL4
MIR155HG
IL5
IL17A
PDCD1
IL23A
STAT5A
CD44
PSAT1
BIRC5
ZDHHC12
ORC1
CDC20
PCNA
EXO1
CDT1
CDC6
EIF4EBP1
CHAC1
TCF19
CDC25A
RFC2
MAPK3
TUBG1
CACYBP
RFC4
Log2 Fold change
0.5 4.0 7.5
Log2 Fold change
-1.0 -2.0 -3.0
B
GSEA-KEGG
Cytokine-cytokine receptor interaction
Viral protein interaction with cytokine and cytokine receptor
Allograft rejection
Asthma
Intestinal immune network for IgA production
Inflammatory bowel disease
Systemic lupus erythematosus
TNF signaling pathway
Natural killer cell mediated cytotoxicity
Chemokine signaling pathway
JAK-STAT signaling pathway
NF-kappa B signaling pathway
T cell receptor signaling pathway
Th17 cell differentiation
Primary immunodeficiency
Th1 and Th2 cell differentiation
C-type lectin receptor signaling pathway
Complement and coagulation cascades
PD-L1 expression and PD-1 checkpoint pathway in cancer
IL-17 signaling pathway
Cellular senescence
Homologous recombination
Base excision repair
Oocyte meiosis
Progesterone-mediated oocyte maturation
Nucleotide excision repair
Fanconi anemia pathway
Mismatch repair
DNA replication
Cell cycle
Immune Activation
Immune Regulatory
Mitosis
-log10(p.adjust)
1 2 3 4 5
NES
1 0 -1 -2
-2 -1 0 1 2
NES
C
CD44
CD27
p = 0.1
NKp44
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PD1
p = 0.06
CCR7
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CD62L
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CD95
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CD127
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GeoMean
Vehicle
DA-100nm
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DA
Vehicle
0.01 μM
0.1 μM
1.0 μM
CD27
CD62L
CD27^L CD62L^h
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CD27^h CD62L^L
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DA 1.0 μM
DA 0.1 μM
DA 0.01 μM
Vechicle
0 50 100
Percentage
E
Vehicle
DA (0.1 μM)
DA (1.0 μM)
# of cytokines
0 1 2 3 4
TNFα
IFNγ
IL2
IL17A
IL4
Vehicle
DA (0.1 μM)
DA (1.0μM)
%
Cytokine number (#)
0 1 2 3 4 5
F
Gene expression fold change (DA / Vehicle)
	HD-1	HD-2	HD-3	HD-4	HD-5	HD-6	HD-7	Pair t test
TCF1*	4.3	2.9	2.8	1.7	1.6	1.1	1.0	0.038
LEF1	1.4	2.0	1.3	1.9	1.3	0.6	0.8	0.132
MYC*	2.2	2.9	4.3	1.6	1.6	1.2	1.8	0.022
T-bet*	2.4	1.8	1.0	1.7	1.3	1.2	1.8	0.015
Eomes	1.0	1.6	1.1	1.6	2.2	2.4	0.9	0.052
MYB	1.1	0.6	0.5	0.9	1.3	2.1	0.8	0.887
ROR	1.3	1.0	2.8	0.3	1.4	0.3	0.6	0.777
GATA3	1.7	1.1	0.8	1.2	1.3	1.4	0.8	0.187
TOX	1.5	0.8	0.5	1.0	1.0	2.2	1.1	0.501

◂**Fig. 3** Dasatinib promotes memory phenotypes and cytokine expression in γδ2 T cells. **A** Heatmap of differentially expressed genes (DEGs) from RNA-seq between dasatinib-treated and control γδ2 T cells; selected cytokines and memory genes highlighted. **B** GSEA-KEGG pathway analysis of upregulated genes. **C** Flow cytometry validation of memory and cytotoxic markers (CD62L, CD127, CCR7, CD95, and NKp44). Paired *t*-test is used, $^{*}p<0.05$, $^{**}p<0.01$, and $^{***}p<0.001$. **D** Increased frequency of $CD62L^{+}$ and $CD62L^{+}CD27^{+}$ subsets in dasatinib-treated γδ2 T cells in different concentrations on the day 7. Average percentage of the subsets from six donors was shown at the bottom. **E** Pie chart of cytokine expression of γδ2 T cells. Intracellular cytokine staining shows elevated TNF-α^{+} and IFN-γ^{+} cells and increased polyfunctionality. Results from six donors were shown as means ± SD (bottom). Student *t*-test is used, $^{*}p<0.05$. **F** Expression of crucial transcription factors of T cell development. TCF1, MYC, and T-bet revealed statistic differences

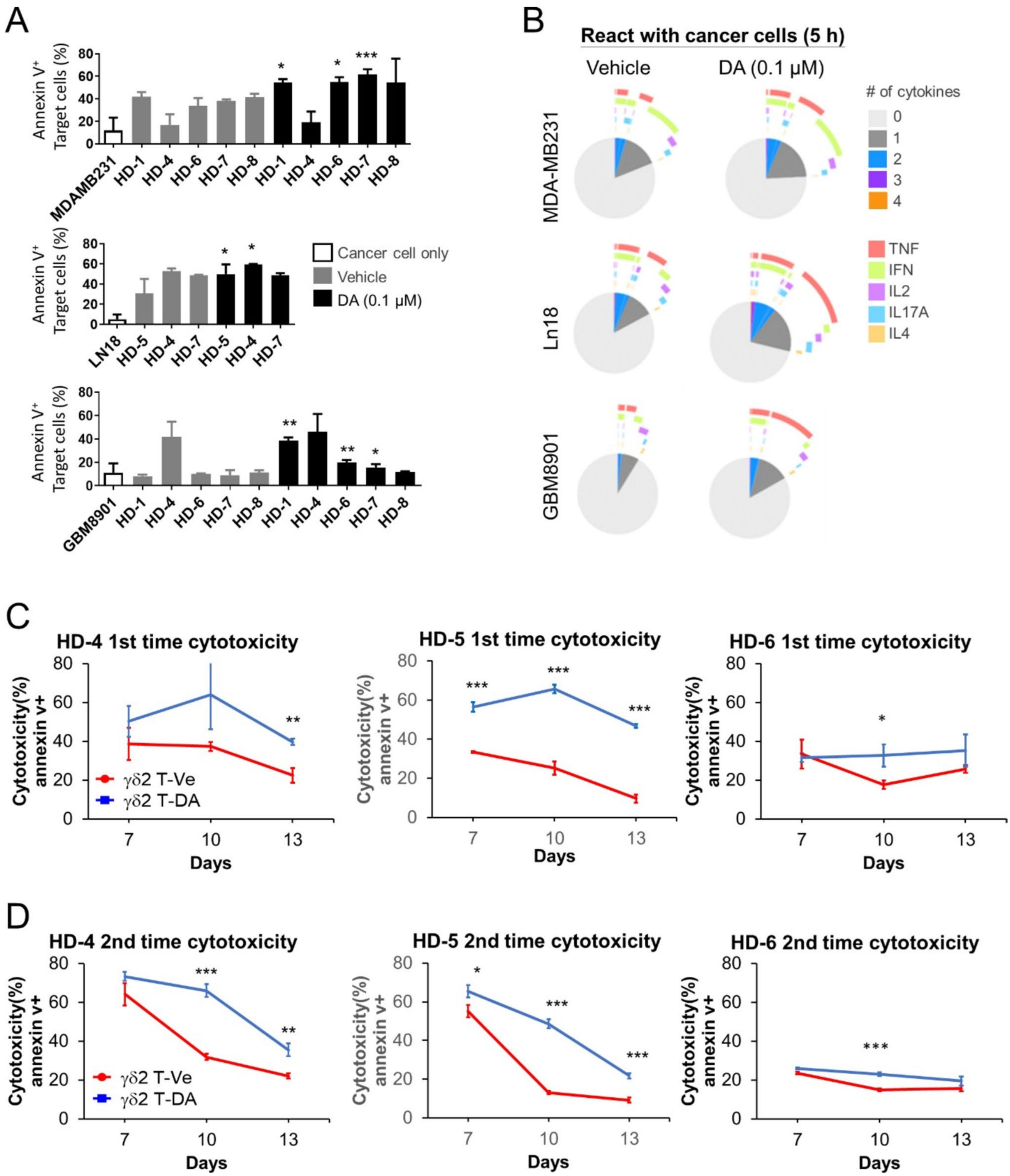
cell proliferation (Fig. 1A). We observed that donors with a high initial percentage of γδ2 cells exhibited superior expansion capacity, reaching 80–90% of total CD3 + T cells (good expanders), whereas donors with a low initial γδ2 proportion only achieved 40–60% γδ2 cell expansion (poor expanders). This result is consistent with previous studies proposing that interindividual heterogeneity in the γδ T cell population, particularly the initial percentage of γδ2 cells in PBMCs, is a key determinant of ex vivo expansion capacity [23]. The $CD3^{+}V\delta2^{+}$ cell population declined from day 0 to day 3 before undergoing a sharp increase, suggesting that PBMC-derived γδ2 T cells contain both terminally differentiated effectors, which survive for only a few days, and memory cell subsets, which are capable of proliferation (Fig. 1B and C).

To comprehensively define the immunophenotypes of γδ2 T cells, we performed multiparameter flow cytometry to assess critical surface markers associated with T cell memory (CD27, CD62L, CD44, CD127, and CCR7), activation (CD95, PD-1), and cytotoxicity (NKp44 and CD56) during ex vivo expansion. We combined CD3 + $V\delta2^{+}$ cells from five healthy donors. t-SNE mapping and FlowSOM clustering identified nine phenotypically distinct populations (P0–P8), with P1 (30%), P3 (35%), and P6 (16%) being the predominant subsets of freshly isolated γδ2 T cells (day 0) (Fig. 1D). After 3 days of Zol-IL2 culture, the total cell counts decreased, and the percentages of P1 and P3 populations declined, while the P6 population expanded to 35%. By day 5, P3 was the dominant subset (77%), whereas P1 gradually increased from day 5 to day 10. Based on the surface marker expression and cluster dynamics, we suggest that P6 comprises central memory γδ2 T cells, characterized by relatively high expression of memory T cell markers, whereas P1 represents terminally differentiated γδ2 T cells, with decreased memory and cytotoxicity markers (Fig. 1E).

Histogram analysis of each marker from day 3 to day 10 revealed a sharp decline in CD44 expression by day 10, while CD62L, CCR7, CD127, and NKp44 expression gradually decreased over the culture period (Fig. 1F). In addition, an in vitro co-culture assay with the Ln18 cancer cell line indicated a gradual decrease in the cytotoxicity of γδ2 T cells against cancer cells when the Zol-IL2 culture was extended from 7 to 13 days (Fig. 1G). These findings suggest that although Zol-based ex vivo expansion can support robust γδ2 T cell proliferation for over 2 weeks, prolonged culture may impair memory and cytotoxicity phenotypes.

In vitro drug screening identifies compounds that promote γδ2 T cell expansion

Accordingly, we suggest that day 3 to day 7 of Zol-IL2 culture is critical for γδ2 T cell expansion and phenotypic changes. To identify compounds that can support proliferation and preserve T cell functionality, we screened an FDA-approved drug library during Zol-IL2 culture. A total of 892 drugs, including antidiabetic, antiseizure, antipsychotic, antibiotic, antiviral, nonsteroidal anti-inflammatory, and chemotherapeutic agents were tested. For the initial screening, each compound (1.0 μM) was added to the culture medium on day 3 (Fig. 2A). Because memory T cells preserve the best proliferation capacity, we first compared the fold changes of absolute γδ2 T cell number relative to the vehicle control from two healthy donors (Fig. 2B). Excluding compounds that induced significant cell death, this screening protocol displayed a positive correlation between γδ2 T cells and the percentage of total CD3 + T cells (Fig. 2C and Table S5), suggesting that γδ2 T cells were the most affected. Notably, the top two γδ2 T cell-promoting drugs dasatinib and ponatinib are tyrosine kinase inhibitors (TKIs) that target the BCR-ABL1 pathway [24–26]. Other TKIs (bosutinib, imatinib, and nilotinib) with similar functions in targeting the BCR-ABL1 pathway also had positive effects on γδ2 T cell expansion; thus, all five were selected for further validation. Among them, dasatinib consistently showed the strongest expansion effects across all tested healthy donors (Fig. 2D, n = 7). Dose–response analysis revealed interindividual variation in γδ2 T cell sensitivity to dasatinib, with 0.1 μM producing significant effects in all donors (Fig. 2E). Importantly, dasatinib treatment increased both the absolute number and percentage of γδ2 T cells in poor expanders, donors with initially low γδ2 T cell frequencies. However, pretreatment or co-treatment with dasatinib in PBMC cultures did not enhance γδ2 T cell expansion (Fig. S1A), suggesting that early-timing administration of dasatinib may not be beneficial for γδ2 T cell expansion in this setting. Assessing proliferation and apoptosis, we found that dasatinib treatment markedly reduced caspase-3/7 activity, while cell division rates remained unchanged (Fig. 2G, S1B), indicating that dasatinib may avoid γδ2 T apoptosis rather than enhance proliferation.



Dasatinib enhances memory phenotypes of γδ2 T cells

To evaluate the phenotypic changes induced by dasatinib, we performed mRNA sequencing of sorted γδ2 T cells cultured with or without dasatinib. PMA/ionomycin stimulation was used to provide uniform downstream activation independent of accessory cell-mediated pAg presentation. Differential gene expression analysis (top 2,000 expressed genes; Table S6) revealed significant upregulation of cytokine genes

◂**Fig. 4** Dasatinib-cultured γδ2 T cells exhibit superior in vitro antitumor activity and persistence. **A** Cytotoxicity of γδ2 T cells expanded with dasatinib (γδ2 T-Da) or vehicle (γδ2 T-Ve) against MDA-MB-231 (TNBC), Ln18, and GBM8901 (GBM) cancer cells in a 5-h co-culture assay. Tumor cell apoptosis was measured by Annexin V positivity. While γδ2 T-Ve cells showed donor-variable cytotoxicity, γδ2 T-Da cells consistently exhibited enhanced tumor cell killing in a majority of tested donors. **B** Polyfunctional cytokine profiling of γδ2 T cells post co-culture, assessed by intracellular staining for TNF-α, IFN-γ, IL2, IL4, and IL17A. γδ2 T-Da cells displayed higher proportions of TNF-α^+ and IFN-γ^+ cells, indicating a more robust effector response. **C** Long-term cytotoxicity against GBM8901 cells using γδ2 T-Da and γδ2 T-Ve cells harvested at day 7, 10, and 13 of ex vivo expansion. γδ2 T-Da cells retained greater cytotoxicity over time, whereas γδ2 T-Ve cells showed a progressive loss of killing ability. **D** Serial killing assay demonstrating that γδ2 T-Da cells maintain cytotoxic function over repeated tumor cell challenges, suggesting enhanced persistence and durability of antitumor activity. Student *t*-test is used, $*p<0.05$, $**p<0.01$ and $***p<0.001$. The effector-to-target (E:T) ratios used in **A**, **C**, and **D** were 1:1

(IL2, IL22, IL4, IL17, and IL5) and memory-associated markers (IL7R/CD127 and CD44) following dasatinib treatment. In addition, several genes related to T cell activation, including XCL2, IFNG, MIR155HG, PDCD1, and STAT5A, were upregulated (Fig. 3A), and MIR155HG was identified as the most significantly upregulated gene (Fig. S2A). In contrast, genes involved in DNA replication and cell cycle regulation, such as PSAT1, CDC20, and PCNA, were downregulated, potentially reflecting the known effects of dasatinib on kinase activity and proliferation.

Gene set enrichment analysis (GSEA) of KEGG pathways revealed that dasatinib-induced differentially expressed genes (DEGs) were positively enriched in immune activation and regulation pathways, whereas mitosis-related pathways were negatively enriched (Figs. 3B, S2B).

Flow cytometry analysis across multiple donors confirmed that dasatinib-treated γδ2 T cells exhibited increased surface expression of NKp44, CD62L, and CD127 along with decreased expression of CCR7 and CD95 (Fig. 3C). Notably, dasatinib promoted a dose-dependent shift toward $CD62L^+$ single-positive and $CD62L^+CD27^+$ double-positive phenotypes (Fig. 3D), consistent with a memory T cell profile associated with reduced apoptosis and enhanced long-term survival. Measurement of cell activity using intracellular cytokine staining after PMA/ionomycin stimulation showed that dasatinib-treated γδ2 T cells had higher proportions of $TNF\alpha^+$ and $IFN\gamma^+$ cells, with an increase in polyfunctional cytokine-producing cells (Fig. 3E). Furthermore, analysis of critical T cell development transcription factors in sorted γδ2 T cells indicated significant upregulation of T-bet and MYC, which are essential for T cell proliferation, as well as the self-renewal regulator TCF1, in the dasatinib-treated group (Fig. 3F). Collectively, these findings suggest that dasatinib enhances γδ2 T cell survival and promotes memory-like phenotypes.

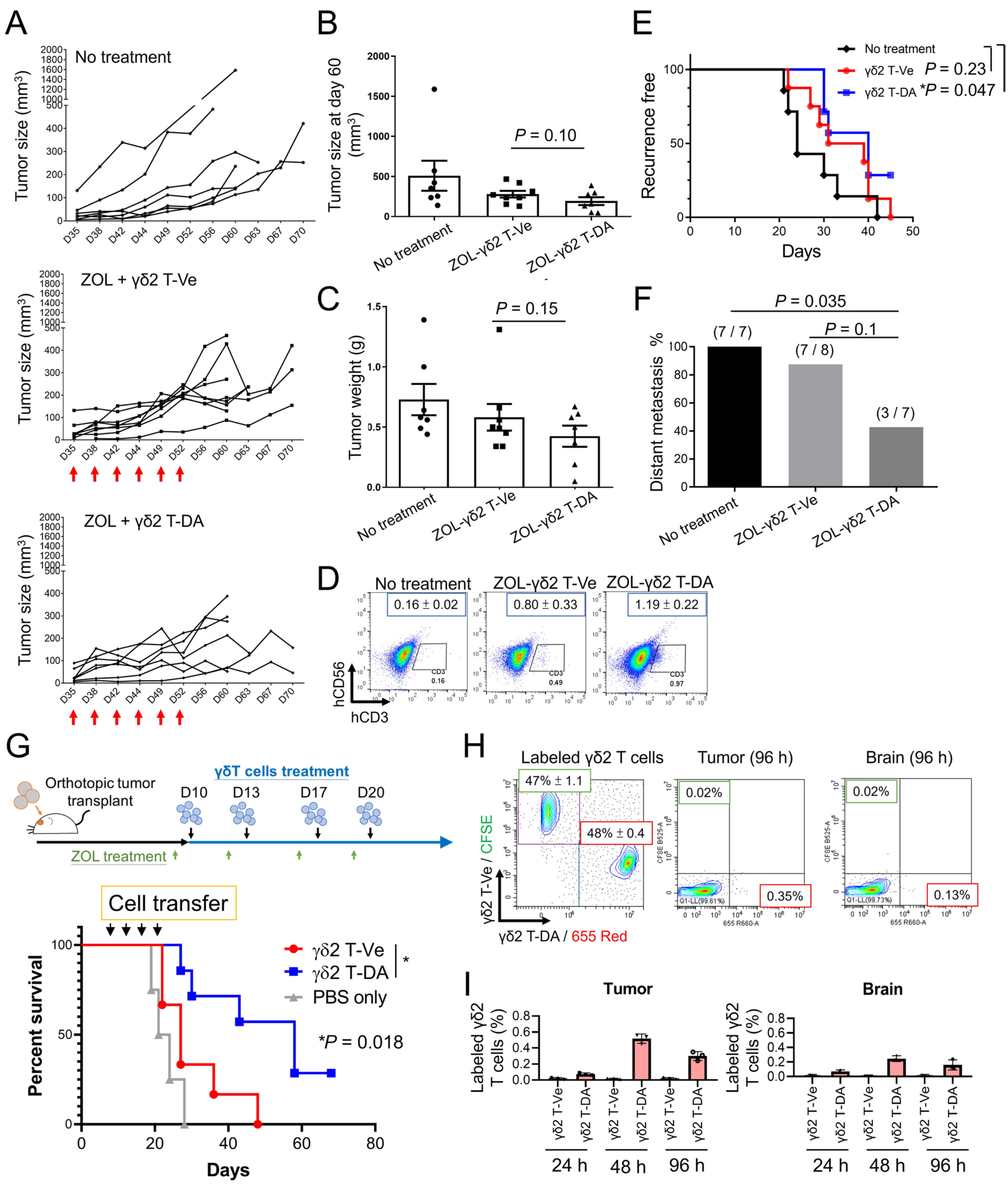
Dasatinib-cultured γδ2 T cells exhibit superior in vitro antitumor activity

Aggressive solid tumors, such as glioblastoma (GBM) and triple-negative breast cancer (TNBC), are notoriously difficult to target with adoptively transferred T cells and represent a significant unmet need for effective curative therapies. Next, we compared the cytotoxicity of dasatinib-treated γδ2 T cells (γδ2 T-Da) and vehicle-treated γδ2 T (γδ2 T-Ve) cells against GBM (Ln18, GBM8901) and TNBC (MDA-MB-231) cells. We first performed a co-culture assay for 5 h and analyzed Annexin V-positive cancer cells. Although there is large individual variation, most donor γδ2 T-Ve cells showed cytotoxicity against MDA-MB231 and Ln18. Only one of the five tested donors showed obvious cytotoxicity to GBM8901 cells within a short time (Fig. 4A). Importantly, roughly 60% of γδ2 T-Da cells showed superior cytotoxicity compared to the paired γδ2 T-Ve control (Fig. 4A). The co-culture with tumor cells increased Ki-67 expression in both vehicle- and dasatinib-treated γδ2 T cells, with no significant difference observed between groups (data not shown). By measuring the cytokine production of co-culture γδ2 T cells by flow cytometry, higher proportions of $TNF\alpha^+$-and $IFN\gamma^+$-expressing cells were found in the dasatinib-treated group, indicating stronger responses to the cancer cell lines (Fig. 4B).

Our findings in Fig. 1 suggest a gradual decrease in memory phenotypes and antitumor activities during extended ex vivo culture of γδ2 T cells. Next, we examined the effect of γδ2 T-Da cells on antitumor activity. The difficult to target GBM8901 cells were co-cultured for 24 h with γδ2 T-Da or γδ2 T-Ve cells that were expanded ex vivo for 7, 10, or 13 days. Our data indicated that γδ2 T-Da cells not only showed better cytotoxicity against cancer cells, but also maintained cytotoxicity after long-term expansion culture (Fig. 4C). Moreover, a serial cancer cell killing assay revealed that γδ2 T-Da cells showed persistent cytotoxicity (Fig. 4D). These results suggest that our established dasatinib culture conditions may have positive effects on the antitumor activity of γδ2 T cells.

γδ2 T-Da cells performed superior antitumor activity in vivo

To evaluate the in vivo antitumor efficacy of γδ2 T-Da cells, we used orthotopic xenograft tumor models in immunodeficient mice. In the triple-negative breast cancer (TNBC) model, MDA-MB-231 cells were transplanted into the mammary fat pad, followed by intratumoral injections of γδ2 T cells from healthy donor 4 (HD-4) twice a week for 3 weeks (Fig. 5A). Although statistical significance was not achieved, γδ2 T-Da cell treatment exhibited a trend toward reduced tumor growth compared with the control groups (Fig. 5B



and C). Following surgical resection of the primary tumor, a higher number of γδ2 T-Da cells was retained within the tumor tissue than in the γδ2 T-Ve-treated group (Fig. 5D). We found that hVδ2⁺ cells constituted the majority of human T cells recovered from tumors (52–60%, data not shown). Because low event numbers at endpoint and sample debris limited reliable dot-plot visualization; therefore, total hCD3⁺ infiltration was used as a consistent surrogate readout of

◂**Fig. 5** γδ2 T-Da cells demonstrate enhanced in vivo antitumor activity and persistence. **A** Orthotopic TNBC model. MDA-MB-231 cells were implanted into the mammary fat pad of immunodeficient mice, followed by intratumoral injection of γδ2 T-Ve or γδ2 T-Da cells twice per week for 3 weeks (red arrow). Tumor growth curves show tumor volume. **B** Tumor size at day 60 and **C** final tumor weight measured after surgical resection. **D** Flow cytometric analysis of $CD3^+$ γδ T cells retained in tumor tissue post-resection, indicating higher tumor infiltration in the γδ2 T-Da group. **E** Kaplan–Meier survival curve of local recurrence-free survival following tumor resection. γδ2 T-Da treatment significantly delayed recurrence. **F** Quantification of distant metastasis frequency. **G** In the GBM model, γδ2 T cells were administered intravenously 10-day post-tumor implantation, preceded by ZOL treatment. γδ2 T-Da cells significantly prolonged survival compared to γδ2 T-Ve controls (log-rank test, $p = 0.018$). **H** Cell trafficking study using fluorescently labeled γδ2 T cells. γδ2 T-Da and γδ2 T-Ve cells were labeled with distinct dyes and transferred into GBM-bearing mice. **I** Flow cytometry showing greater distribution of γδ2 T-Da cells in tumor tissue, while γδ2 T-Ve cells were minimally detectable (n = 3 per group for each time point)

intratumoral T cell persistence. Additionally, the γδ2 T-Da treatment group demonstrated a significantly longer local recurrence-free interval than the untreated group (Fig. 5E) and a reduced incidence of distant metastasis (Fig. 5F).

In a more aggressive glioblastoma (GBM) model, γδ2 T cells from healthy donor 1 (HD-1) were administered intravenously 10 days after tumor implantation. Zoledronic acid (ZOL) was administered 24 h before cell infusion to enhance phosphoantigen expression. Mice treated with γδ2 T-Da cells exhibited significantly improved survival compared to controls (log-rank test, $p = 0.018$; Fig. 5G).

To evaluate T cell persistence and tissue homing, γδ2 T-Ve and γδ2 T-Da cells HD-1 were labeled with distinct fluorescent tracers and adoptively transferred into a new cohort of GBM tumor-bearing mice 10 days after intracranial implantation of GBM8901 cells (Fig. 5H). The purity of Vδ2 T cells is around 90%. At 24-h post-transfer, γδ2 T-Da cells were more prevalent in the spleen than in the bone marrow, and were subsequently detected in the tumor tissues and lymph nodes (Fig. 5I and Supplementary Fig. S3). In contrast, very few γδ2 T-Ve cells were detected in these tissues. Collectively, these findings support that dasatinib-expanded γδ2 T cells possess enhanced antitumor capabilities, potentially due to improve in vivo survival and tumor trafficking.

Discussion

Despite the promising potential of γδ T cells in adoptive immunotherapy, particularly for MHC-independent tumor recognition and innate-like properties, their clinical application remains limited by suboptimal ex vivo expansion protocols and diminished functional persistence [1, 27, 28]. In this study, we systematically characterized the phenotypic dynamics of Vγ9Vδ2 (γδ2) T cells using a commonly used Zol and IL-2-based expansion approach and identified dasatinib, a clinically approved tyrosine kinase inhibitor, as a potent enhancer of γδ2 T cell expansion, memory phenotype retention, and antitumor function.

Tyrosine kinase inhibitors (TKIs) targeting BCR-ABL and SRC oncoproteins, including imatinib, nilotinib, dasatinib, bosutinib, and ponatinib, have been approved for the treatment of chronic myeloid leukemia (CML) and in certain contexts, acute myeloid leukemia (AML) [29–31]. Each TKI has a distinct pharmacological and safety profile. In addition to their direct antitumor effects, growing evidence suggests that TKIs can also modulate the immune system, making them promising candidates as immunomodulatory agents in combination with immunotherapies [32]. For instance, dasatinib induces peripheral expansion of T cells and natural killer (NK) cells in patients [33]. Moreover, studies in CML patients who achieve major or deep molecular responses during TKI therapy have reported increased frequencies of γδ T cells, particularly within the naïve subset [34]. These findings support the dual role of dasatinib as an anticancer agent and immune modulator. Although dasatinib broadly inhibits T cell activation and differentiation programs, raising concerns about potential immunosuppression, it has also been reported to enhance the function of specific immune cell populations, such as NK cells and CD8 + T cells, under defined conditions [25, 35]. This underscores the complex immunological effects that are dependent on cell type and context.

Although all five TKIs targeting BCR-ABL were tested in this study, dasatinib exhibited the most potent effects on γδ T cell expansion and upregulation of memory-associated markers, suggesting its unique immunomodulatory properties. Mechanistically, several factors may have contributed to the superiority of dasatinib. First, inhibition of BCR-ABL by dasatinib can enhance the efflux of intracellular isopentenyl pyrophosphate (IPP), a phosphoantigen that activates Vγ9Vδ2 T cells [36]. Second, dasatinib ideally acts as a reversible inhibitor of CAR-T cell signaling, serving as a pharmacological "pause switch" to transiently suppress T cell activity and cytokine release, revealing its potential for transiently preserving T cell activity [37, 38]. Third, dasatinib possesses well-characterized senolytic properties that selectively induce apoptosis in senescent pre-adipocytes, endothelial cells, and mesenchymal stem cells. This effect could partly be mediated through the inhibition of Src family kinases, such as c-Kit, which are critical for senescent cell survival [39, 40]. Although the precise mechanisms of action of dasatinib in our γδ T cell culture system require further investigation, it is plausible that both the suppression of hyperactive TCR signaling and the removal of senescent cells contribute to the enhanced expansion and functionality observed in our study.

Ibrutinib, a Bruton's tyrosine kinase (BTK) inhibitor, was ranked among the top 11 compounds that promoted γδ2 T cell expansion in our 892-drug screening (Table S5). Interestingly, previous studies have shown that ibrutinib can improve the fitness of Vγ9Vδ2 T cells in patients with chronic lymphocytic leukemia (CLL), likely through its dual inhibition of interleukin-2–inducible kinase (ITK) and BTK, thereby promoting a Th1-skewed phenotype [41, 42]. Furthermore, ibrutinib has been proposed to reverse the functional impairment of Vγ9Vδ2 T cells in CLL and has also been shown to enhance chimeric antigen receptor (CAR) T cell engraftment and antileukemic efficacy [42, 43]. These findings suggest that further optimization of ibrutinib, alone or in combination with dasatinib, within our γδ2 T cell expansion protocol could be a promising strategy worth exploring in future studies.

We found that prolonged Zol-IL2 expansion compromises γδ2 T cell memory markers (e.g., CD127 and CD62L) and cytotoxicity, confirming previous concerns that extended culture induces terminal differentiation or exhaustion. After screening 892 FDA-approved compounds, dasatinib emerged as the most consistent enhancer of γδ2 T cell expansion across multiple donors, especially among poverty-poor expanders. Mechanistically, dasatinib treatment suppressed apoptosis without accelerating proliferation, suggesting a survival-enhancing effect, rather than mitogenic stimulation.

Interestingly, our data also suggest that miR-155, a microRNA known to regulate T cell memory, inflammation, and antitumor responses, is most significantly induced by dasatinib [44–46]. This could partly explain the observed enhancement in memory phenotypes and effector function, as miR-155 modulates T cell survival through SOCS1/SHIP1 repression and supports IFNγ and TNF production [47]. Further investigation of the miR-155 axis in γδ2 T cells may reveal new targets for fine-tuning their therapeutic potential.

In our breast tumor model, intratumoral administration produced the most consistent antitumor responses. Nevertheless, intravenous infusion remains the clinically preferred route and will be prioritized in future translational studies. A known limitation of non-antigen-specific adoptive cell therapies, including Vγ9Vδ2 T cells and NK cells, is inefficient trafficking to solid tumors following systemic delivery. To partially overcome this obstacle, we employed retro-orbital venous injection combined with zoledronate (Zol) preconditioning in our GBM model to enhance γδ2 T cell access to intracranial tumors. This strategy likely improves therapeutic efficacy by increasing BTN2A1/BTN3A1-dependent phosphoantigen presentation, thereby promoting local activation of γδ2 T cells within the tumor microenvironment. [12–16]. We also acknowledge that the use of severely immunodeficient mice limits our ability to fully assess systemic immune dynamics and cell–immune cross-talk, and future studies using humanized or partially reconstituted immune models will be required to more accurately evaluate clinical relevance.

In conclusion, our study demonstrates that dasatinib not only enhances γδ2 T cell expansion but also preserves memory-like phenotypes and cytotoxic potential, providing a practical strategy to improve γδ T cell-based immunotherapy. Although the present work was conducted using γδ T cells from healthy donors, the fact that dasatinib is already FDA-approved offers a near-term opportunity for clinical translation, particularly for ex vivo expansion of autologous γδ2 T cells from patients with solid tumors. Collectively, these findings provide a strong rationale for future clinical studies evaluating dasatinib-augmented γδ T cell products in solid tumor settings.

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Author contributions JR Lin, YC Song, YL Chou, and CH Liu designed the study. YL Chou, YJ Chen, and JP Li performed experiments. Hsiao YJ Ho and WC Liao performed data analyses. HR Yen and CH Liu produced the text of the manuscript and provided guidance for the project.

Data availability The data that support the findings of this study are available from the corresponding author, Chiung-Hui Liu, upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical approval This study was approved by the Ethical Committees of China Medical University Hospital, Taiwan.

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